

Section B- Practical Coding Task

Task 1: Component Identification & Functions

Present your findings in a structured hardware table:

Section B - Task 1: Desktop Core Hardware Components		
Component Identification, Key Hardware Specifications, and Core Operational Functions		
Component	Make / Model & Key Specifications	Primary System Function
Motherboard Foundational PCB	Make / Model: ASUS PRIME B650M-A AX II Form Factor: Micro-ATX (24.4 cm x 24.4 cm) Socket / Chipset: Socket AM5 (LGA 1718) AMD B650 Expansion & I/O: 1x PCIe 4.0 x16, 2x M.2 NVMe, 4x DDR5	Acts as the central backbone circuit board of the PC. It provides physical sockets and electrical pathways (bus lines) to interconnect the CPU, RAM, GPU, storage drives, and peripheral ports, coordinating data communication and distributing DC power.
CPU (Central Processor) Computing Core	Make / Model: AMD Ryzen 5 7600X Desktop Processor Cores / Threads: 6 Cores / 12 Threads (TSMC 5nm FinFET) Frequencies: 4.7 GHz Base Clock / 5.3 GHz Max Boost Cache / TDP: 32 MB L3 Cache 105W TDP Socket AM5	The primary computing 'brain' of the computer. It executes operating system instructions, processes arithmetic logic and program algorithms, and coordinates data movement across all connected hardware subsystems through continuous fetch-decode-execute cycles.
RAM (System Memory) Volatile Storage	Make / Model: Kingston FURY Beast DDR5 Desktop Memory Capacity / Kit: 16 GB (2 x 8 GB) Dual-Channel Kit Speed Rating: DDR5-5200 MT/s (PC5-41600) CL36 Form / Voltage: 288-pin UDIMM 1.25V Intel XMP 3.0 / EXPO	Provides high-speed, volatile short-term working storage directly accessible by the CPU. It holds active operating system files, running software applications, and loaded data sets for near-instantaneous read/write operations; clears completely when power is removed.
GPU (Graphics Card) Graphics & Video	Make / Model: NVIDIA GeForce RTX 4060 Dual OC VRAM Memory: 8 GB GDDR6 (128-bit Bus 272 GB/s) CUDA Cores / Bus: 4302 CUDA Cores PCIe 4.0 x8 Video Outputs: 3x DisplayPort 1.4a, 1x HDMI 2.1a	Accelerates graphical computation and renders 2D/3D imagery. Offloads matrix and vector calculations from the CPU to render geometries, textures, and video frames, transmitting pixel display signals directly to the monitor via HDMI and DisplayPort connections.
Storage Device (M.2 NVMe SSD) Permanent Storage	Make / Model: Crucial P3 Plus M.2 PCIe 4.0 NVMe SSD Capacity & Form: 1TB Storage M.2 2280 Form Factor Interface / Protocol: PCIe Gen 4.0 x4 NVMe 1.4 Protocol Sequential Speeds: Up to 5,000 MB/s Read 4,200 MB/s Write	Provides high-speed, non-volatile long-term storage that permanently retains the OS, applications, and user files when powered off. Utilizes PCIe 4.0 lanes for high IOPS, delivering boot times under 10 seconds and eliminating application loading bottlenecks.
Cooling System (CPU Air Cooler) Thermal Regulation	Make / Model: DeepCool AG400 Single-Tower CPU Cooler Heatsink Design: Aluminum Fin Stack + 4 Direct-Touch Cu Heatpipes Cooling Fan: 120mm PWM Fluid Dynamic Bearing Fan (500-2000 RPM) Thermal Rating: Up to 220W TDP Cooling Capacity	Draws thermal energy away from the CPU integrated heat spreader (IHS) via copper heatpipes and 6 aluminum fins, where the PWM fan exhausts heat outside the chassis. Prevents thermal throttling and hardware degradation under continuous processing loads.

Task 2: Motherboard Ports & Socket Identification

- 1. Major Motherboard Ports & Headers:
 - **24-Pin ATX Main Power:** Supplies primary DC voltages (+3.3V, +5V, +12V) to the motherboard logic board.
 - **8-Pin (4+4) EPS CPU Power:** Located near the top-left; delivers dedicated 12V power to the CPU voltage regulator modules (VRMs).
 - **SATA 6 Gbps Ports:** L-shaped data connectors used for 2.5-inch SATA SSDs or 3.5-inch mechanical hard drives.
 - **Front Panel Headers:** Connects case power switches, reset buttons, power LEDs, and drive activity LEDs.
 - **Internal USB Headers:** Internal pins for front-case USB 3.2 Gen 1 (19-pin) and USB 2.0 (9-pin).
- 2. Expansion Slots & Purpose:
 - **PCIe x16 Slot:** Primary reinforced expansion slot connecting high-bandwidth dedicated graphics cards (GPUs).
 - **PCIe x1 Slots:** Smaller peripheral slots for Wi-Fi expansion cards, sound cards, or capture cards.
 - **M.2 NVMe Slot (Key M):** High-speed slot mounted flat on the board for PCIe Gen3/Gen4 NVMe solid-state drives.
- 3. CPU Socket Type & Compatibility:
 - **Socket Type:** **AMD Socket AM5 (LGA 1718)** (or **Intel LGA 1700** depending on your test bench).
 - **Compatible CPU Families:** AMD Ryzen 7000, 8000, and 9000 desktop series processors (LGA 1700 supports Intel 12th, 13th, and 14th Gen Core i3/i5/i7/i9).

- 4. Diagram / Visual Evidence:

Draw or submit a labeled schematic diagram pointing with arrows to the CPU Socket, RAM DIMM slots, PCIe x16 slot, M.2 slot, 24-pin power connector, and rear I/O shield.

Task 3: RAM & Storage Installation

A. RAM Module Removal & Reinstallation

1. **ESD Precautions:** Unplug the power supply, ground yourself by wearing an anti-static wrist strap clipped to bare metal, and press the power button for 10 seconds to drain residual capacitor charge.
2. **Removal:** Press down on the locking latches at the ends of the DIMM slot until the RAM stick unseats and pops up slightly, then lift it straight out by its edges without touching the golden pins.
3. **Reinstallation:** Align the off-center notch on the bottom of the RAM module with the key in the DIMM slot. Press straight down on both ends with even pressure until both retention latches snap inward with an audible click.

B. Storage Device Removal & Reinstallation (M.2 NVMe)

1. **Removal:** Unscrew the M.2 retaining screw or release the plastic toolless latch. Gently pull the SSD out of the M.2 slot at a 30-degree angle.
2. **Reinstallation:** Slide the M.2 drive firmly into the M-key slot at a 30-degree angle until fully inserted, press it flat against the standoff, and secure it with the mounting screw.
3. **Verification:** Power on the PC, tap **DEL** or **F2** to enter UEFI/BIOS, and verify that the RAM capacity (e.g., 16 GB Dual-Channel) and the exact drive model appear under the **Main / System Information** dashboard.

Task 4: Full Desktop PC Assembly

Follow the standard sequential build order, documenting each step:

1. Pre-Case Preparation (Out-of-Case Assembly):

- Place the motherboard on top of its non-conductive cardboard box (never on top of an anti-static bag).
- Open the CPU retention arm, align the gold triangle on the CPU with the socket marker, drop it into the socket with zero insertion force, and lock the lever down.
- Install the M.2 NVMe SSD into the primary M.2 slot under the heatsink.
- Insert RAM sticks into slots A2 and B2 until they click.
- Apply thermal paste (pea-sized dot) and secure the CPU cooler to the mounting brackets.

2. Case Installation:

- Install the motherboard I/O shield (if not pre-installed) and verify motherboard standoffs align with case screw holes.
- Lower the motherboard into the chassis and secure it with screws.

3. Power Supply & Cabling:

- Mount the PSU into the bottom chamber.
- Route and plug in the **24-pin ATX power cable**, **8-pin EPS CPU cable**, and case front panel headers (Power SW, Reset SW, Front USB, Front Audio).

4. GPU Installation:

- Remove the rear PCIe expansion slot covers, insert the graphics card into the primary PCIe x16 slot until the lock engages, screw it to the chassis bracket, and plug in the dedicated 8-pin PCIe power cable.
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5. Post-Build Validation & First Boot:

- Connect power, plug the HDMI/DisplayPort cable directly into the GPU, and power on the system.
- Confirm the CPU and case fans spin, motherboard debug LEDs cycle through CPU →DRAM →VGA →BOOT, and the display successfully shows the initial BIOS POST screen.